

Question ID: 6ae1360d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A circle has a radius of **2.1** inches. The area of the circle is **$b\pi$** square inches, where **b** is a constant. What is the value of **b** ?

Correct Answer: 4.41, 441/100

Rationale

The correct answer is **4.41**. The area, **A** , of a circle is given by the formula **$A = \pi r^2$** , where **r** is the radius of the circle. It's given that the area of the circle is **$b\pi$** square inches, where **b** is a constant, and the radius of the circle is **2.1** inches. Substituting **$b\pi$** for **A** and **2.1** for **r** in the formula **$A = \pi r^2$** yields **$b\pi = \pi(2.1^2)$** . Dividing both sides of this equation by **π** yields **$b = 4.41$** . Therefore, the value of **b** is **4.41**.